

JEE-Main-06-04-2024 (Memory Based)
[MORNING SHIFT]

Physics

Question: Which of the following can't be explained by wave nature of Light?

Options:

- (a) Diffraction
- (b) Photoelectric effect
- (c) Interference
- (d) Polarisation

Answer: (b)

Question: When a dc supply of 100 V is applied across an inductor then the current passes through it is 5A. Find the power across it in an AC source of peak voltage 200 V is applied across it given inductive reactance is $20\sqrt{3} \Omega$.

Options:

- (a) 100 W
- (b) 200 W
- (c) 250 W
- (d) 500 W

Answer: (c)

Question: Find the maximum speed in the case of an SHM where time period and amplitude are 3.14 sec and 0.06 m respectively.

Options:

- (a) 0.06 m/s
- (b) 3.14 m/s
- (c) 0.12 m/s
- (d) 0.24 m/s

Answer: (c)

Question: A bullet is fired which passes through a plywood. Find the loss in KE of the 50 g bullet if it enters at 100 m/s and exits at 40 m/s.

Options:

- (a) 150 J
- (b) 210 J
- (c) 480 J
- (d) 500 J

Answer: (b)

Question: Match the following.

Physical quantities	Dimensional formula
i) Torque	P) $M^1L^0T^{-2}A^{-1}$
ii) Magnetic moment	Q) $M^1L^2T^{-1}$
iii) Magnetic field	R) $M^1L^2T^{-2}$
iv) Angular momentum	S) $M^0L^2T^0A^1$

Options:

- (a) i-R; ii-S; iii-P; iv-Q
- (b) i-S; ii-P; iii-Q; iv-R
- (c) i-P; ii-Q; iii-R; iv-S
- (d) i-P; ii-R; iii-S; iv-Q

Answer: (a)

Question: Particles of mass $m/2$, m , $2m$, $4m$ are having same momentum then which of the particle will have maximum kinetic energy?

Options:

- (a) Particle of mass $4m$
- (b) Particle of mass $2m$
- (c) Particle of mass m
- (d) Particle of mass $m/2$

Answer: (d)

Question: Minimum kinetic energy require for a body of mass m to go to infinity from earth's surface, is

Options:

- (a) $\frac{GMm}{2R}$
- (b) $\frac{GMm}{R}$
- (c) $\frac{2GMm}{R}$
- (d) Zero

Answer: (b)

Question: What is the expression of electric field on the surface of a uniformly charged spherical shell having surface charge density as σ and radius R ?

Options:

- (a) $\frac{\sigma R}{\epsilon_0}$
- (b) $\frac{\sigma}{2\epsilon_0}$
- (c) $\frac{\sigma}{\epsilon_0}$
- (d) $\frac{\sigma}{4\epsilon_0}$

Answer: (c)

Question: Find ratio of V_{RMS} of Helium & Oxygen at same temperature

Options:

- (a) $\sqrt{2}$
- (b) $2\sqrt{2}$
- (c) 2
- (d) $\sqrt{3}$

Answer: (b)

Question: A soap bubble of radius R is split into 1000 small bubbles. The ratio of surface energy of 1000 bubbles to the big bubble is

Options:

- (a) 100 : 1
- (b) 1 : 1
- (c) 10 : 1
- (d) 1 : 100

Answer: (c)

Question: Find ratio of Shortest wavelengths of Balmer series & Lyman Series

Options:

- (a) 4 : 1
- (b) 3 : 1
- (c) 2 : 1
- (d) 1 : 1

Answer: (a)

Question: Energy of incident photon is 2.48 eV and stopping potential 0.5 V. Find work function.

Options:

- (a) 1 eV
- (b) 2.98 eV
- (c) 2.5 eV
- (d) 1.98 eV

Answer: (d)

Question: A train is accelerating from rest with a constant acceleration “a” for a time t sec and attain a speed of 80 m/s, then for a time of $3t$ it runs with a constant speed. Find the average speed.

Options:

- (a) 70 m/s
- (b) 35 m/s
- (c) 80 m/s
- (d) 20 m/s

Answer: (a)

Question: A wire of resistance is stretched to a radius of $r/2$. Where 'r' is the initial radius. The new resistance of the wire is

Options:

- (a) $4R$
- (b) $16R$
- (c) $R/4$
- (d) $R/16$

Answer: (b)

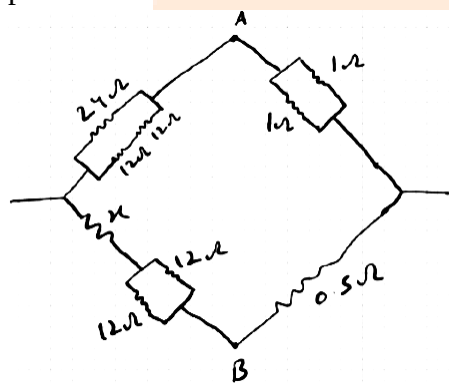
Question: An infinite long wire having 10 A current passing through it along x axis. Find the magnetic field at 0.5 m above the origin

Options:

- (a) $4\pi \times 10^{-6} T$
- (b) $\frac{1}{4\pi} \times 10^{-6} T$
- (c) $4 \times 10^{-6} T$
- (d) $2 \times 10^{-6} T$

Answer: (c)

Question: Find the resistance x in figure, if it is provided that points A and B are at same potential

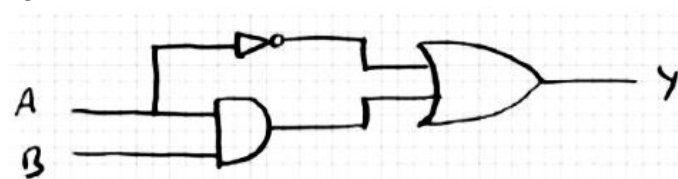


Options:

- (a) 4Ω
- (b) 12Ω
- (c) 6Ω
- (d) 8Ω

Answer: (c)

Question: Which will be the correct truth table for following circuit diagram?



Options:

(a)

A	B	Y
1	1	0
1	0	1
0	1	1
0	0	1

(b)

A	B	Y
1	1	1
1	0	0
0	1	1
0	0	1

(c)

A	B	Y
1	1	1
1	0	0
0	1	0
0	0	1

(d)

A	B	Y
1	1	0
1	0	1
0	1	1
0	0	0

Answer: (b)

Question: An electromagnetic wave travels in a medium with the speed 1.5×10^8 m/s. The relative permeability of medium is 2. Find the relative permittivity.

Options:

(a) 1

(b) $\frac{1}{2}$

(c) 2

(d) $\frac{3}{2}$

Answer: (c)

Question: If the ratio of height & angle of prism is 1 & refractive index is $\sqrt{3}$ then find the value of angle of prism?

Options:

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{6}$

(d) $\frac{\pi}{3}$

Answer: (d)

Question: Two masses M_1 & M_2 , where $M_2 > M_1$ are connected by a string which passes over a massless and frictionless pulley.

The masses accelerate at $\frac{g}{\sqrt{2}}$ m/s² in opposite directions, then find the ratio of M_1 & M_2

Answer: $3 - 2\sqrt{2}$

Question: The diameter of a wire is measured using a screw gauge of 100 division and pitch 1mm. Main scale reading is 1mm, circular scale reading is 42. Diameter is found to be $x/50$ mm then x is

Answer: 71

Question: A current carry coil of 100 turns, area 10 cm^2 , current $5 \times 10^{-5} \text{ A}$ is placed in a magnetic field of strength 1T. Initially plane of the coil is perpendicular to the magnetic field. The work done in rotating the coil by 90° is $x \times 10^{-6} \text{ J}$ then find x.

Answer: 5



JEE-Main-06-04-2024 (Memory Based)
[MORNING SHIFT]

Chemistry

Question: Density of NaOH is 1.12g/mL having molality of 3m. Calculate the molarity.

Options:

- (a) 3 M
- (b) 2 M
- (c) 1 M
- (d) 0.5 M

Answer: (a)

Question: Which of the following not semiconductor ?

Options:

- (a) Gallium
- (b) Copper oxide
- (c) Graphite
- (d) Silicon

Answer: (c)

Question: $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} \xrightarrow{\text{H}^+} ?$

Options:

- (a) CO_2 , MnO_4^{2-}
- (b) CO_2 , MnO_4^+
- (c) CO_2 , Mn^{2+}
- (d) CO_2 , Mn^{7+}

Answer: (c)

Question: Among the given molecules, identify the one which undergoes nucleophilic addition reaction at fastest rate

Options:

- (a) HCHO
- (b) CH_3CHO
- (c) $\text{CH}_3\text{CH}_2\text{CHO}$
- (d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$

Answer: (a)

Question: Find out ratio of $t_{99.9\%}$ and $t_{90\%}$ for first order ?

Options:

- (a) 2

- (b) 1
(c) 3
(d) 4

Answer: (c)

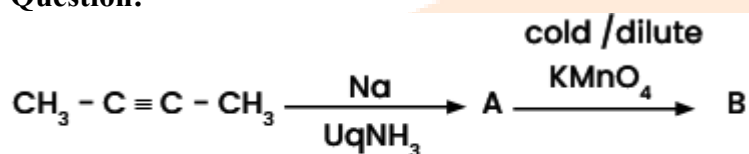
Question: Find the ratio of shortest wavelengths in Lyman and Balmer series for H-atom.

Options:

- (a) 1/4
(b) 4
(c) 1/2
(d) 2

Answer: (a)

Question:



Number of oxygen atom is product B ?

Options:

- (a) 4
(b) 3
(c) 1
(d) 2

Answer: (d)

Question: Which compound will absorb light at more frequency?

Options:

- (a) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
(b) $[\text{CrCl}_6]^{3-}$
(c) $[\text{Cr}(\text{CN})_6]^{3-}$
(d) $[\text{CrCl}_3(\text{H}_2\text{O})_3]$

Answer: (c)

Question: Structure Based Questions

- | | |
|---------------------|----------------|
| a. SF_4 | 1. Tetrahedral |
| b. BrF_3 | 2. Pyramidal |
| c. BrO_3^- | 3. Sea-saw |
| d. NH_4^+ | 4. T-Shape |

Options:

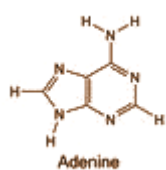
- (a) (a) → 4, (b) → 2, (c) → 1, (d) → 3
(b) (a) → 3, (b) → 4, (c) → 2, (d) → 1
(c) (a) → 1, (b) → 3, (c) → 2, (d) → 4
(d) (a) → 4, (b) → 2, (c) → 1, (d) → 3

Answer: (b)

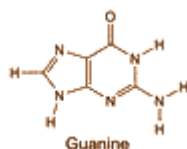
Question: Identify which of the base is not present in DNA-

Options:

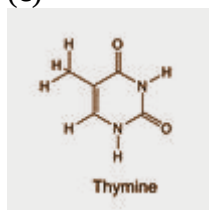
(a)



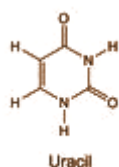
(b)



(c)



(d)



Answer: (d)

Question: Reimer - Tiemann reaction involves as intermediate :

Options:

- (a) Carbocation
- (b) Carbanion
- (c) Carbanion
- (d) Carbene

Answer: (d)

Question: Match the following :-

- | | |
|-------------------------|----------------------|
| a. Iodoform | 1. Fire extinguisher |
| b. DDT | 2. Insecticide |
| c. Carbon Tetrachloride | 3. Antiseptic |
| d. Chlorofluorocarbon | 4. Refrigerator |

Options:

- (a) (a) → 3, (b) → 2, (c) → 1, (d) → 4
- (b) (a) → 2, (b) → 3, (c) → 4, (d) → 1
- (c) (a) → 1, (b) → 3, (c) → 2, (d) → 4
- (d) (a) → 4, (b) → 3, (c) → 1, (d) → 2

Answer: (a)

Question: Find functional group present in sulphuric acid

Options:

- (a) $-\text{NO}_2$
- (b) $-\text{SO}_2$
- (c) $-\text{SO}_3\text{H}$
- (d) $-\text{COOH}$

Answer: (c)

Question: Match the following

Hybridisation

Structure

(P) sp^3d^2

(A) Octahedral

(Q) sp^3

(B) Trigonal Bipyramidal

(R) dsp^2

(C) Tetrahedral

(S) sp^3d

(D) Square planar

Options:

- (a) $\text{P} \rightarrow \text{A}$, $\text{Q} \rightarrow \text{C}$, $\text{R} \rightarrow \text{D}$, $\text{S} \rightarrow \text{B}$
- (b) $\text{P} \rightarrow \text{B}$, $\text{Q} \rightarrow \text{A}$, $\text{R} \rightarrow \text{C}$, $\text{S} \rightarrow \text{D}$
- (c) $\text{P} \rightarrow \text{B}$, $\text{Q} \rightarrow \text{D}$, $\text{R} \rightarrow \text{A}$, $\text{S} \rightarrow \text{C}$
- (d) $\text{P} \rightarrow \text{C}$, $\text{Q} \rightarrow \text{A}$, $\text{R} \rightarrow \text{D}$, $\text{S} \rightarrow \text{B}$

Answer: (a)

Question: Statement I:- 2, 4, 6-trinitrotoluene is known as picric acid

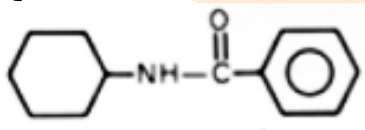
Statement II:- Phenol can be converted into picric acid by addition of concentrated HNO_3 in phenol-2,4-disulphonic acid.

Options:

- (a) S1-Correct ; S2-Incorrect
- (b) S1-Incorrect ; S2-Correct
- (c) S1-Correct ; S2-Correct
- (d) S1-Incorrect ; S2-Incorrect

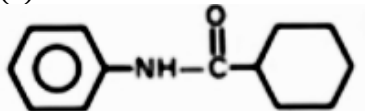
Answer: (b)

Question: Which one is correct metamer of

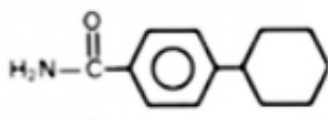


Options:

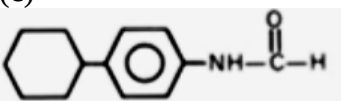
(a)



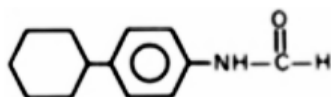
(b)



(c)



(d)



Answer: (a)

Question: Match the following :

- | | |
|------------------------|---|
| (i) Pb^{2+} | (a) $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$ |
| (ii) Al^{3+} | (b) $\text{H}_2\text{S} + \text{dil HCl}$ |
| (iii) Sr^{2+} | (c) $\text{H}_2\text{S} + \text{NH}_4\text{OH}$ |
| (iv) Mn^{2+} | (d) $(\text{NH}_4)_2\text{CO}_3 + \text{NH}_4\text{OH}$ |

Options:

- (a) (i) $\rightarrow$ (a), (ii) $\rightarrow$ (b), (iii) $\rightarrow$ (d), (iv) $\rightarrow$ (c)
 (b) (i) $\rightarrow$ (a), (ii) $\rightarrow$ (c), (iii) $\rightarrow$ (b), (iv) $\rightarrow$ (d)
 (c) (i) $\rightarrow$ (b), (ii) $\rightarrow$ (a), (iii) $\rightarrow$ (d), (iv) $\rightarrow$ (c)
 (d) (i) $\rightarrow$ (b), (ii) $\rightarrow$ (d), (iii) $\rightarrow$ (a), (iv) $\rightarrow$ (c)

Answer: (c)

Question: Which of the following are element of lanthanide series. Eu, Cm, Cr, Yb, Lu, Cd, Tb, Er.

Options:

- (a) 4
 (b) 3
 (c) 5
 (d) 2

Answer: (c)

Question: Which of the following have negative electron affinity?

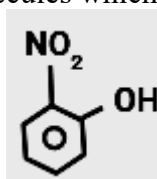
- a. $\text{B} \longrightarrow \text{Be}^-$
 b. $\text{N} \longrightarrow \text{N}^-$
 c. $\text{O} \longrightarrow \text{O}^{2-}$
 d. $\text{Na} \longrightarrow \text{Na}^-$
 e. $\text{Al} \longrightarrow \text{Al}^-$

Options:

- (a) b, c, d
 (b) b, c, d
 (c) c, d, e
 (d) b, c, f

Answer: (b)

Question: Number of molecules which can show H-bonding among



CH_3OH , H_2O , C_2H_6 , C_6H_6 , HF , NH_3

Options:

- (a) 5
 (b) 4

(c) 6

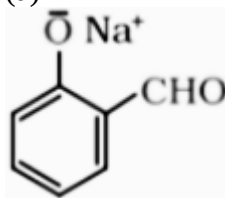
(d) 3

Answer: (a)

Question: Which of the following is not the intermediate observed in Reimer-Tiemann reaction

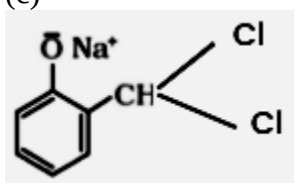
Options:

(a)



(b) :CCl_2

(c)



(d) CHCl_3

Answer: (d)

Question: Find the sum of magnetic moment of basic and amphoteric oxides of Cr. CrO , Cr_2O_3 CrO_3

Answer: 8.76 BM

JEE-Main-06-04-2024 (Memory Based)
[MORNING SHIFT]

Maths

Question: No. of triangles that can be formed from a regular octagon, provided no. side of octagon should be a side of triangle.

Options:

- (a) 8
- (b) 12
- (c) 14
- (d) 16

Answer: (d)

Question: Let the area of the region enclosed by curves $y = 3x$, $2y = 27 - 3x$ and $y = 3x - 2\sqrt{x}$ be A. Then, $10A$ is equal to 162, 184, 154, 172

Options:

- (a) 122
- (b) 132
- (c) 152
- (d) 162

Answer: (d)

Question: Find interval in which x^x is strictly increasing

Options:

- (a) $(0, \infty)$
- (b) $\left(0, \frac{1}{e}\right]$
- (c) $\left[\frac{1}{e^2}, \infty\right)$
- (d) $\left(\frac{1}{e}, \infty\right)$

Answer: (d)

Question: Two factories A and B. 60% cars were made in A factory and remaining were made in B factory. Then we have 80% cars from Factory A is the standard Quality 90% of cars from factory B is a standard quality. A car is picked randomly and found it as standard, the probability that car came from B is P. Find $126P$?

Options:

- (a) 54
- (b) 52

(c) 48

(d) 27

Answer: (a)

Question: Let a circle touch the parabola $y = 6 - x^2$ and touch the lines $y = \sqrt{3}|x|$ such that the circle has minimum area. Then which of the following points lie on the circle

Options:

(a) (2, 2)

(b) (1, 1)

(c) (1, 2)

(d) (2, 4)

Answer: (d)

Question: If $A_r = \begin{vmatrix} r & 1 & \frac{\pi^2}{2} + \alpha \\ 2r & 2 & n^2 - \beta \\ 3r - 1 & 3 & \frac{n}{2}(3n - 1) \end{vmatrix}$ then the value of $2A_{10} - A_8$ is equal to

Options:

(a) $4\alpha + 2\beta$

(b) $2n$

(c) 0

(d) $2\alpha + 4\beta$

Answer: (a)

Question: If

$$f(x) = \begin{cases} x^3 \sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0 \end{cases} \text{ then}$$

Options:

(a) $f''\left(\frac{2}{\pi}\right) = \frac{12 - \pi^2}{2\pi}$

(b) $f'(0) = 0$

(c) $f'(0) = 1$

(d) $f''\left(\frac{2}{\pi}\right) = \frac{24 - \pi^2}{2\pi}$

Answer: (a)

Question: R is defined on set $X = \{1, 2, \dots, 20\}$ and $R_1 = \{(x, y) : 2x - 3y = 2\}$, $R_2 = \{(x, y) : 5x - 4y = 0\}$. If M, N represent the number of elements to be added to make R_1 & R_2 symmetric respectively. Then find the value of $M + N$.

Options:

(a) 10

(b) 8

(c) 12

(d) 11

Answer: (a)

Question: If mean and standard deviation of 20 observations are 10 and 2. It was better found that one of the value was 8 instead of 12. Find the correct standard deviation.

Options:

(a) 1.8

- (b) $\sqrt{3.96}$
 (c) $\sqrt{3.84}$
 (d) 1.93

Answer: (b)

Question: The value of $\int_0^{\frac{\pi}{4}} \frac{\cos^2 x \sin^2 x}{(\cos^3 x + \sin^3 x)^2} dx$ is equal to

Options:

- (a) $\frac{1}{6}$
 (b) $\frac{1}{3}$
 (c) $\frac{1}{2}$
 (d) 1

Answer: (a)

Question: Let $y = y(x)$ be the solution of the differential equation $(2x \log_e x) \frac{dy}{dx} + 2y = \frac{3}{x} \log_e x, x > 0$ and $y(e-1) = 0$ the $y(e)$ is equal to

Options:

- (a) $\frac{-3}{e}$
 (b) $\frac{-3}{3e}$
 (c) $\frac{-3}{2e}$
 (d) $\frac{-2}{e}$

Answer: (a)

Question: If the function $f(x) = \frac{x^2 + 2x - 15}{x^2 - 4x + 9}; x \in R$ is

Options:

- (a) Neither one-one - nor onto
 (b) One-one but not onto
 (c) Onto but not one-one
 (d) Both one-one and onto

Answer: (a)

Question:

$$x^2 - (t^2 - 5t + 6)x + 1 = 0$$

$$A_n = \alpha^n + \beta^n$$

Find min value $\frac{A_{2025} + A_{2023}}{A_{2024}}$

Answer: (-1/4)

Question: If $2^{\text{nd}}, 3^{\text{rd}}, 4^{\text{th}}$ term of $(x + y)^n$ are respectively 135, 30, $10/3$. Find $6[n^3 + x^2 + y]$

Answer: 806